

Package & Yarn Dyeing Machinery Constraints Manual

1. Overview of Package Dyeing Dynamics

In yarn package dyeing, spooled yarn packages are loaded onto perforated dye spindles inside a pressurized cylindrical vessel. Dye liquor is forced radially through yarn package walls under high pump pressure.

2. Target Package Densities by Fiber Type

Fiber Substrate	Target Density (g/cm³)	Permissible Var (±)	Hardness (Shore A)
Cotton (Spun Yarn)	0.32 - 0.38 g/cm³	± 0.015 g/cm³	45 - 55
100% Polyester (Textured)	0.38 - 0.42 g/cm³	± 0.010 g/cm³	55 - 65
Wool & Wool Blends	0.28 - 0.32 g/cm³	± 0.015 g/cm³	35 - 45
Acrylic (High Bulk)	0.26 - 0.30 g/cm³	± 0.010 g/cm³	30 - 40

3. Reversal Schedules & Differential Pressure

Cotton (Reactive): In-Out 4 min / Out-In 2 min.

Polyester (Disperse): In-Out 5 min / Out-In 3 min.

Differential Pressure Delta P: 0.8 bar to 1.4 bar (Low P < 0.5 bar indicates bypass; High P > 1.8 bar indicates package clogging).